

FillFinder: how it works

A step by step walkthrough on a simple flower design. Prototype v2, Datix AI.

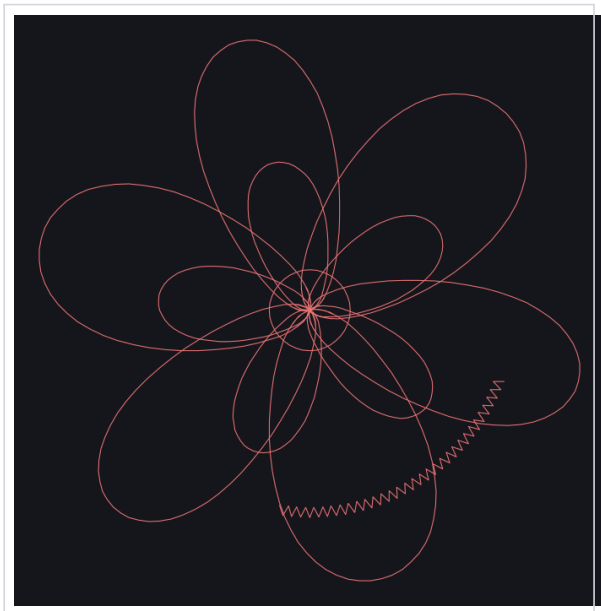
The problem

A digitizer draws a design with outline stitches. The lines cross each other and overlap, so the software sees no closed shapes and cannot fill anything automatically. Every petal has to be traced by hand with Complex Fill and then reshaped to match the outline. One neckline design can contain 70 or more of these shapes, which is one to three hours of repetitive work.

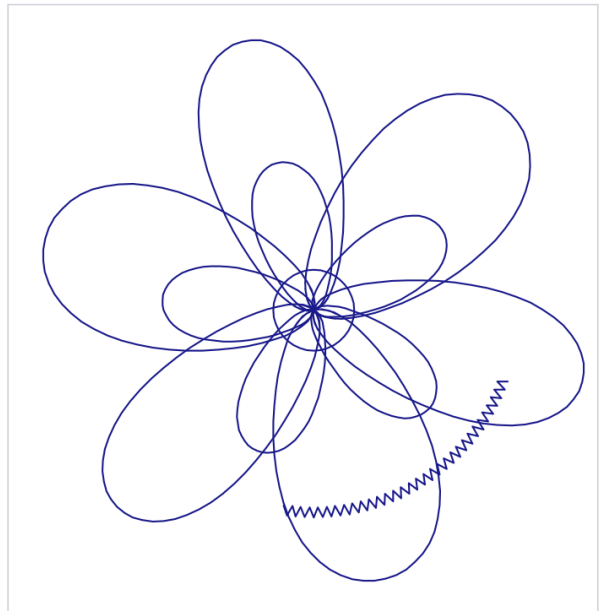
FillFinder does that detection automatically in a few seconds.

Step 1: the input

The example is a flower drawn only with open running lines. No petal is a closed object: each line runs past its own start so the ends cross, petals overlap each other, one petal is left with a small gap on purpose, and a zigzag border runs across the bottom. This is exactly what the client's screen shows before he starts filling.



Input file: 774 stitches, one colour, open lines only



Every line is kept and used as an outline

Step 2: lines become walls

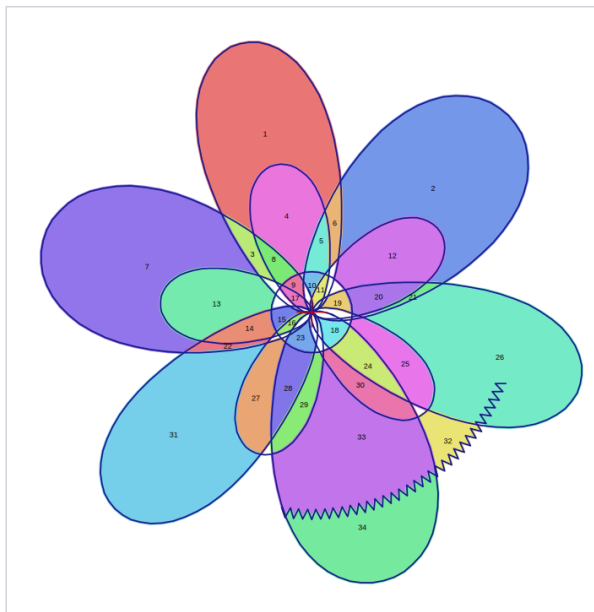
Each stitch line is thickened slightly, by default into a strip 0.6 mm wide. This one step solves three problems at once: gaps smaller than 0.6 mm are sealed, a zigzag or satin border turns into one solid band, and crossing lines join into a connected network.

Step 3: enclosed areas are detected

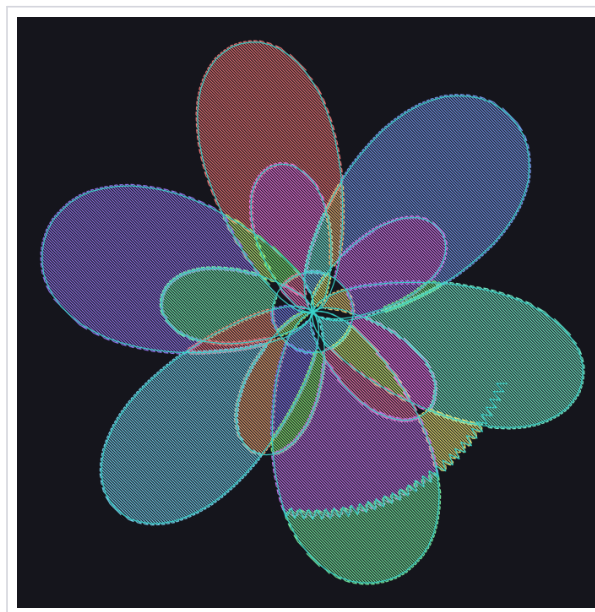
Any space completely surrounded by those walls is a fillable area. Areas smaller than 4 mm² are dropped as slivers, and anything as large as the background is skipped. Each area is then grown back slightly so the fill tucks under the outline and no fabric shows at the edge.

On this flower it found **34 areas** in under half a second, including areas that exist only because two petals overlap, and the petal that was drawn with a gap.

Step 4: the result



34 areas detected and numbered



Fill stitches generated for every area

What the tool saves

	By hand today	With FillFinder
Finding the shapes	trace each petal manually	automatic, a few seconds
Matching the outline	drag reshape points	follows the lines exactly
Gaps in the outline	close them by hand	closed automatically
Shapes in this example	34 traced one by one	34 found at once

Step 5: what you get from each run

Every run is saved in its own dated folder and nothing is overwritten:

report.html a one page before and after summary

regions_for_wilcom.svg the filled shapes at the exact same position and scale, ready to import into Wilcom

autofilled.dst and **.pes** a new stitch file for preview

regions.csv and **summary.json** area sizes, positions and settings used

four SVG previews original, outlines kept, areas found, filled result

How it is meant to be used in production

The fill stitches in the prototype are preview quality: one fixed angle, no underlay and no pull compensation. The production route is to import **regions_for_wilcom.svg** into Wilcom, where the shapes arrive as closed objects at the right position. The digitizer then applies whatever stitch type he wants, Tatami, Satin or radial, and Wilcom handles angles, underlay and quality.

Planned next: fill angle that follows each petal, radial fill for round flowers, original colours per area, and a standalone Windows application.